TUTORIAL 4

Linear Systems, Gaussian Elimination, and Linear Combinations

4.1 Notation

NOTATION AND CONVENTIONS

Let F be a field. The set F^n consists of all columns $(x_1, \ldots, x_n)^T$ with $x_i \in F$. Here T indicates column notation. Addition and scalar multiplication are componentwise. The set $M_{m\times n}(F)$ consists of matrices with m rows and n columns over F. The entry in row i and column j is a_{ij}. For a system with n unknowns and m equations, append the column of right-hand sides to the coefficient matrix to obtain the augmented matrix.

MOTIVATION AND GUIDING IDEA

Row reduction replaces a problem with an equivalent one that is easier to read. After calculating, return to the question: is the system consistent, which variables are free, and what is the solution set? For a nonhomogeneous system, the difference of two solutions solves the associated homogeneous system. This explains the form ``a particular solution plus a homogeneous solution.''

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4.2 Linear systems and row reduction

4.2.1 A system of linear equations

RECALL AND TOOLS

In general, for a positive integer n, a linear equation in n unknowns (variables) has the form

$$a_1x_1 + a_2x_2 + \cdots + a_nx_n = b$$

Here $a_1, a_2, ..., a_n$ and b are **given scalars** in a specified field $\mathbb{F}$, such as $\mathbb{R}$, $\mathbb{C}$, or a finite field, and $x_1, x_2, ..., x_n$ are the **unknowns**, or variables.
The scalar a_i is the **coefficient** of x_i, and b is the **constant term** of the equation.

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A system of m equations in n unknowns has the general form

$$\begin{cases} a_{1,1}x_1 + a_{1,2}x_2 + \cdots + a_{1,n}x_n = b_1 \\ a_{2,1}x_1 + a_{2,2}x_2 + \cdots + a_{2,n}x_n = b_2 \\ \vdots \\ a_{m,1}x_1 + a_{m,2}x_2 + \cdots + a_{m,n}x_n = b_m \end{cases}$$

To solve the system, it is convenient to collect the **coefficients** in a matrix, preserving the variable order. To shorten the notation, we write either the matrix of variable coefficients, called the **coefficient matrix**, or the **augmented matrix**, which also includes the constant terms. Following the basic rules consistently makes it clear which coefficient belongs to each variable.

EXAMPLE 4.1

$$\begin{cases} 3x_1 - 2x_2 = 0, \\ -6x_1 + 5x_2 = 2 \end{cases} \longleftrightarrow \left(\begin{array}{cc|c} 3 & -2 & 0 \\ -6 & 5 & 2 \end{array}\right).$$

4.2.2 Elementary operations

DEFINITION 4.1

Elementary row operations:
An elementary row operation modifies a matrix in one of the following ways:

1. Interchange rows i and j, denoted by $R_i \longleftrightarrow R_j$.
2. Multiply row i by a **nonzero** scalar λ, denoted by $R_i \longleftarrow \lambda R_i$.
   (Dividing by a nonzero scalar a is multiplication by $1/a$.)
3. Add λ times row j to a different row i, where $i \neq j$, denoted by $R_i \longleftarrow R_i + \lambda R_j$.

- Elementary row operations preserve the solutions of a linear system.
- Every elementary row operation has an inverse that is also an elementary row operation.

DEFINITION 4.2

Matrices A and B are **row equivalent** (or simply equivalent), denoted by $A \sim B$, if B is obtained by applying a finite sequence of elementary row operations to A.

DEFINITION 4.3

Leading entry (pivot): in every nonzero row of a matrix, the first nonzero entry is called its leading entry.

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DEFINITION 4.4

Row echelon form: each leading entry is strictly to the left of the leading entry in the next nonzero row. Any zero rows appear below all nonzero rows.

DEFINITION 4.5

Reduced row echelon form: a matrix in row echelon form such that, in each nonzero row, the leading entry equals 1, and every other entry above and below it in the same column equals 0. These requirements apply to every column, including the constant column of an augmented matrix.

PROPOSITION 4.1

Every matrix is row equivalent to a unique matrix in reduced row echelon form.

Examples:

1. $\begin{pmatrix} 2 & 1 \\ 3 & -1 \end{pmatrix}$ is not in echelon form: its leading entries are in the same column.
2. $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 3 \end{pmatrix}$ is in echelon form but not reduced echelon form.
3. $\begin{pmatrix} 1 & 0 & -2 \\ 0 & 1 & 4 \end{pmatrix}$ is in reduced row echelon form.

4.2.3 Gaussian elimination

RECALL AND TOOLS

1. Start with the first active row and the first column. Look in that column for a nonzero entry in the active row or below it. If none exists, move to the next column.
2. Interchange rows to bring this entry into the active row. Division by it is permitted because it has been checked to be nonzero.
3. Multiply the active row by the inverse of its leading entry to obtain a leading 1, and use it to eliminate the entries below.
4. Move to the next active row and seek a pivot column to the right of the previous one. A pivot column need not have the same index as its row.
5. After reaching row echelon form, eliminate the entries above each pivot, working from the last pivot to the first, to obtain reduced row echelon form. This completes Gauss-Jordan elimination.

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EXERCISE 4.1

Please reduce the augmented matrix over $\mathbb{R}$

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 2 & 1 & 3 & 12 \\ 1 & 3 & 2 & 3 \end{array}\right)$$

to reduced row echelon form and please find the system's solution.

WORKED SOLUTION

Apply $R_2 \leftarrow R_2 - 2R_1$ and $R_3 \leftarrow R_3 - R_1$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & -3 & 1 & 10 \\ 0 & 1 & 1 & 2 \end{array}\right).$$

Swap the last two rows and apply $R_3 \leftarrow R_3 + 3R_2$:

$$\left(\begin{array}{ccc|c} 1 & 2 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 4 & 16 \end{array}\right).$$

Divide R_3 by 4, subtract it from R_2, and clear the entries above the last two pivots:

$$\left(\begin{array}{ccc|c} 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & -2 \\ 0 & 0 & 1 & 4 \end{array}\right).$$

The unique solution is $(1, -2, 4)^T$. Substitution gives $1-4+4 = 1$, $2-2+12 = 12$, $1-6+8 = 3$.

Another one down. Victory!

NOTE

To solve a linear system, reduce its **augmented matrix**, including the constant terms. You may stop at row echelon form and use back substitution, or continue to reduced row echelon form and read off the solutions.

EXAMPLE 4.2

Consider the system

$$\begin{cases} x + 2y = 0, \\ y + z = 2, \\ 2x + z = 7. \end{cases}$$

Its augmented matrix is

$$\left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 2 & 0 & 1 & 7 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 2 \\ 0 & 0 & 5 & 15 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 0 & 2 \\ 0 & 1 & 0 & -1 \\ 0 & 0 & 1 & 3 \end{array}\right).$$

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The first step uses $R_3 \leftarrow R_3 - 2R_1 + 4R_2$. Hence $z = 3$, $y = 2 - z = -1$ and $x = -2y = 2$. The solution set is $\{(2, -1, 3)\}$.

This example shows how row reduction helps solve a linear system. We will next see that a system need not have a unique solution.

EXERCISE 4.2

Please reduce the augmented matrix over $\mathbb{Z}_{11}$

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 2 & 3 & 4 & 1 & 5 \end{array}\right)$$

to reduced row echelon form. Please perform the arithmetic modulo 11.

WORKED SOLUTION

Apply $R_3 \leftarrow R_3 - 2R_1 - 3R_2$. The last row has entries $4-4-9 = -9 \equiv 2$, $1-2-6 = -7 \equiv 4$ and $5 - 6 - 12 = -13 \equiv 9$. Thus

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 0 & 0 & 2 & 4 & 9 \end{array}\right).$$

The inverse of 2 is 6. Multiplying R_3 by 6 gives

$$\left(\begin{array}{cccc|c} 1 & 0 & 2 & 1 & 3 \\ 0 & 1 & 3 & 2 & 4 \\ 0 & 0 & 1 & 2 & 10 \end{array}\right).$$

Finally apply $R_1 \leftarrow R_1 - 2R_3$ and $R_2 \leftarrow R_2 - 3R_3$:

$$\left(\begin{array}{cccc|c} 1 & 0 & 0 & 8 & 5 \\ 0 & 1 & 0 & 7 & 7 \\ 0 & 0 & 1 & 2 & 10 \end{array}\right).$$

All entries use representatives $0, 1, \ldots, 10$.

We've got it. Victory!

EXERCISE 4.3

Please solve the following system over $\mathbb{C}$:

$$\begin{cases} iz + (2 - i)w = -3 - i, \\ 2iz + (5 - 2i)w = -5 - 5i. \end{cases}$$

Please verify the solution in both equations.

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EXERCISE 4.3 CONTINUED

WORKED SOLUTION

Subtract twice the first equation from the second to get

$$w = (-5 - 5i) - 2(-3 - i) = 1 - 3i.$$

Now $(2 - i)(1 - 3i) = -1 - 7i$, so

$$iz = -3 - i - (-1 - 7i) = -2 + 6i, \qquad z = 6 + 2i.$$

The first equation gives $(-2 + 6i) + (-1 - 7i) = -3 - i$. The second gives

$$2iz + (5 - 2i)w = (-4 + 12i) + (-1 - 17i) = -5 - 5i.$$

That does it. Victory!

4.3 Solving linear systems

DEFINITION 4.6

After row reduction, a variable whose column in the coefficient matrix contains a leading entry is called a **pivot variable** (or dependent variable). A variable whose column contains no leading entry is called a **free variable**. In a consistent system, every free variable can be assigned an arbitrary value in the field, after which the pivot variables are determined by the equations. Over an infinite field, a free parameter gives infinitely many choices; over a finite field, the number of choices is finite. The number of free variables is the number of degrees of freedom of a consistent system.

DEFINITION 4.7

The rank of a matrix A, denoted by $\text{rank}(A)$, is the number of nonzero rows in a row echelon form. Equivalently, it is the number of leading entries (pivots) in that form.

Remark: if a system with coefficient matrix A and n variables is consistent, its number of degrees of freedom is $n - \text{rank}(A)$. Over $\mathbb{F}_q$, it has exactly $q^{n-\text{rank}(A)}$ solutions.

4.4 Linear Combinations and Spanning in F^n

DEFINITION 4.8

For $v_1, \ldots, v_k \in F^n$, a linear combination is a column $\alpha_1 v_1 + \cdots + \alpha_k v_k$, where $\alpha_j \in F$. The set of all such combinations is their span, denoted $\text{span}\{v_1, \ldots, v_k\}$. The list spans F^n if every column in F^n is such a combination. The span of the empty list is $\{0\}$, where 0 is the zero column.

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EXAMPLE 4.3

For $u = (1,1,0)^T$ and $v = (0,1,1)^T$ over a field F,

$$\alpha u + \beta v = (\alpha, \alpha+\beta, \beta)^T.$$

Their span therefore consists of the columns $(x,y,z)^T$ satisfying $y = x + z$. To test whether a given column belongs to the span, solve for α, β.

EXERCISE 4.4

Please solve over $\mathbb{R}$ the system

$$\begin{cases} x + 3y - 2z = 7, \\ 2x - y + 3z = 0, \\ 4x + 5y - z = 14. \end{cases}$$

WORKED SOLUTION

Write the augmented matrix and subtract multiples of its first row:

$$\left(\begin{array}{ccc|c} 1 & 3 & -2 & 7 \\ 2 & -1 & 3 & 0 \\ 4 & 5 & -1 & 14 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 3 & -2 & 7 \\ 0 & -7 & 7 & -14 \\ 0 & -7 & 7 & -14 \end{array}\right).$$

Subtract row two from row three, divide row two by -7 and clear its pivot column:

$$\left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 0 & 1 & -1 & 2 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Put $z = t$. Then $x = 1 - t$, $y = 2 + t$, so

$$S = (1,2,0) + \operatorname{span}\{(-1,1,1)\}.$$

There is one free variable and infinitely many solutions. Substitution gives $7, 0, 14$ in the three equations for every $t \in \mathbb{R}$.

All wrapped up. Victory!

EXERCISE 4.5

Please solve the system

$$\begin{cases} 2x_1 - 6x_2 + 4x_3 = 7, \\ x_1 - 3x_2 + 2x_3 = 2. \end{cases}$$

WORKED SOLUTION

Swap the rows, then subtract twice row one from row two:

$$\left(\begin{array}{ccc|c} 2 & -6 & 4 & 7 \\ 1 & -3 & 2 & 2 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -3 & 2 & 2 \\ 0 & 0 & 0 & 3 \end{array}\right).$$

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EXERCISE 4.5 CONTINUED

The last row says $0 = 3$, so there is no solution.

We've reached the finish. Victory!

Geometric illustration

We have encountered three possible cases:

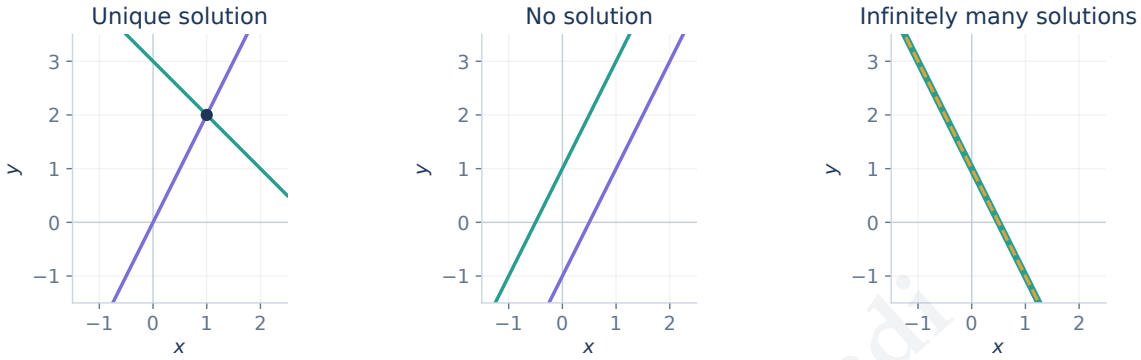


Figure 4.1. Over the real numbers, two equations in two unknowns can describe intersecting lines, distinct parallel lines, or coincident lines. These give, respectively, a unique solution, no solution, or infinitely many solutions.

EXERCISE 4.6

Let $a \in \mathbb{R}$. Please classify the number of solutions and please state the coefficient matrix rank in each case:

$$\begin{cases} x - 2y = 0, \\ 3x + 3z = 2, \\ x + az = 1. \end{cases}$$

WORKED SOLUTION

Subtract three times row one from row two, and row one from row three. Swap the last two rows and subtract three times row two from row three:

$$\left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 3 & 0 & 3 & 2 \\ 1 & 0 & a & 1 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 2 & a & 1 \\ 0 & 0 & 3-3a & -1 \end{array}\right).$$

If $a = 1$, there is a contradiction row and rank 2. If $a \neq 1$, the rank is 3 and the solution is unique:

$$z = -\frac{1}{3-3a}, \qquad x = 1 + \frac{a}{3-3a}, \qquad y = \frac{1}{2}\left(1 + \frac{a}{3-3a}\right).$$

No value of a gives infinitely many solutions.

Yes! Victory!

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4.4.1 Homogeneous systems

RECALL AND TOOLS

All constant terms are 0. In this case:

- We may write only the coefficient matrix rather than the augmented matrix: the zero column is unchanged by all row operations, so there is no need to keep writing it.
- A solution always exists: no contradiction row can occur, so inconsistency is impossible. We must still determine whether the zero solution is the only solution. Over an infinite field, the existence of a nonzero solution implies infinitely many solutions.
- If there is a unique solution, it is the zero solution, also called the **trivial solution**.
- The solution set of a homogeneous system in n variables is a vector subspace of $\mathbb{F}^n$ and can always be expressed as a span.

EXERCISE 4.7

Let $a \in \mathbb{R}$. Please classify the number of solutions of this homogeneous system and please find the solution set in each case:

$$\begin{cases} x - 2y = 0, \\ 3x + 3z = 0, \\ x + az = 0. \end{cases}$$

WORKED SOLUTION

The coefficient matrix is the same as in the preceding exercise. Row reduction gives

$$\begin{pmatrix} 1 & -2 & 0 \\ 3 & 0 & 3 \\ 1 & 0 & a \end{pmatrix} \sim \begin{pmatrix} 1 & -2 & 0 \\ 0 & 2 & a \\ 0 & 0 & 3-3a \end{pmatrix}.$$

If $a \neq 1$, the rank is 3 and only the zero solution exists. If $a = 1$, the rank is 2, and $x = -z$, $y = -z/2$. Choosing $z = 2t$ gives

$$S = \{(-2t, -t, 2t) : t \in \mathbb{R}\} = \text{span}\{(-2, -1, 2)\}.$$

There are infinitely many solutions in this case. A homogeneous system is always consistent, so no case has zero solutions.

V-I-C-T-O-R-Y!

4.5 Supplementary exercises: measurements and counting solutions

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EXERCISE 4.8

Consider the system

$$x_1 + x_2 = 7, \qquad x_2 + x_3 = 11, \qquad x_1 + x_3 = 8.$$

Please find x_1, x_2, x_3. Please find the only value of c for which adding $x_1 + x_2 + x_3 = c$ leaves the solution set unchanged. Please determine what happens for $c = 14$.

WORKED SOLUTION

Add the first two equations and subtract the third:

$$2x_2 = 7 + 11 - 8 = 10, \qquad x_2 = 5.$$

Then $x_1 = 7 - 5 = 2$ and $x_3 = 11 - 5 = 6$. The third equation checks as $2 + 6 = 8$. Adding all three equations gives

$$2(x_1 + x_2 + x_3) = 7 + 11 + 8 = 26.$$

Thus the required value is $c = 13$. Setting $c = 14$ creates a contradiction, so the augmented system has no solution.

Victory!

EXERCISE 4.9

Please solve over $\mathbb{Z}_3$ the system

$$\begin{cases} x + y + z = 1, \\ x + z = 2. \end{cases}$$

Please count its solutions and please compare with the same equations over $\mathbb{R}$.

WORKED SOLUTION

Subtracting the second equation from the first gives $y = -1 = 2$ in $\mathbb{Z}_3$. Choosing $z = t$ gives

$$(x, y, z) = (2 - t, 2, t), \qquad t \in \mathbb{Z}_3.$$

The three solutions are $(2, 2, 0), (1, 2, 1), (0, 2, 2)$. Over $\mathbb{R}$ we get

$$(x, y, z) = (2 - t, -1, t), \qquad t \in \mathbb{R},$$

so there are infinitely many solutions. Both systems have one free variable, but the number of solutions depends on the field size.

And we're done. Victory!

Additional practice

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EXERCISE 4.10

Please find a spanning set for each solution space over $\mathbb{R}$.

1. $\begin{cases} x + 2y = 0, \\ x - z = 0. \end{cases}$
2. $2y - z = 0$.

WORKED SOLUTION

1. Setting $y = t$ gives $x = -2t$ and $z = x = -2t$. Hence

$$S_1 = \{t(-2, 1, -2) : t \in \mathbb{R}\} = \text{span}\{(-2, 1, -2)\}.$$

   Substitution verifies that each multiple satisfies both equations.

2. Set $x = s$, $y = t$; then $z = 2t$. Thus

$$(x, y, z) = s(1, 0, 0) + t(0, 1, 2), \qquad S_2 = \text{span}\{(1, 0, 0), (0, 1, 2)\}.$$

   Both generators satisfy the equation, and the representation includes every solution.

You know what that means? V-I-C-T-O-R-Y!

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EXERCISE 4.11 CHALLENGE

Consider a finite linear system over a field F with $n \geq 1$ unknowns. Assume every collection of at most $n+1$ of its equations is consistent. Please prove that the full system is consistent. Please construct an inconsistent system of $n+1$ equations for which every collection of n equations is consistent.

WORKED SOLUTION

Suppose the system is inconsistent and choose an inconsistent collection of minimum size k. Write its equations as

$$\sum_{i=1}^{n} a_{ji} x_i = b_j, \qquad 1 \leq j \leq k.$$

The hypothesis gives $k > n+1$. Consider the homogeneous system in $\lambda_1, \ldots, \lambda_k$ given by

$$\sum_{j=1}^{k} \lambda_j a_{ji} = 0 \quad (1 \leq i \leq n), \qquad \sum_{j=1}^{k} \lambda_j b_j = 0.$$

It has $n+1$ equations and more than $n+1$ unknowns. Row reduction therefore gives a nonzero solution. Choose j with $\lambda_j \neq 0$. These identities express equation j, including its right-hand side, as a linear combination of the other equations. Hence every solution of the other equations also satisfies equation j. Minimality of k says that the other equations have a solution. That solution satisfies the whole collection, a contradiction.
The bound is sharp. Take

$$x_1 = 0, \ldots, x_n = 0, \qquad x_1 + \cdots + x_n = 1.$$

The full system is inconsistent. If the final equation is omitted, zero is a solution. If $x_j = 0$ is omitted, setting $x_j = 1$ and all other unknowns to zero gives a solution. Thus every collection of n equations is consistent.

Another one down. Victory!

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